

Lecture 9:

Hierarchical Bayesian Modeling

September 23, 2026

HMC sampling using NUTS in JAX

ast384c/inclass/lec8_hmc/hmc_nuts.ipynb

we now know how to sample high dimension distributions:
rejection sampling, importance sampling, MCMC, Hamiltonian MC

let's apply it to astro! choose your adventure:

ast384c/inclass/lec9_hierarchical/review_emission.ipynb

ast384c/inclass/lec9_hierarchical/review_sed.ipynb

there will be additional time to work on these next week

Hierarchical Bayesian Modeling

often in astro, **population inference** is the science

galaxy evolution: *luminosity function, stellar mass function*

cosmology: *photometric redshift distribution, cluster mass function*

time-domain: *GRB luminosity function, transient rates*

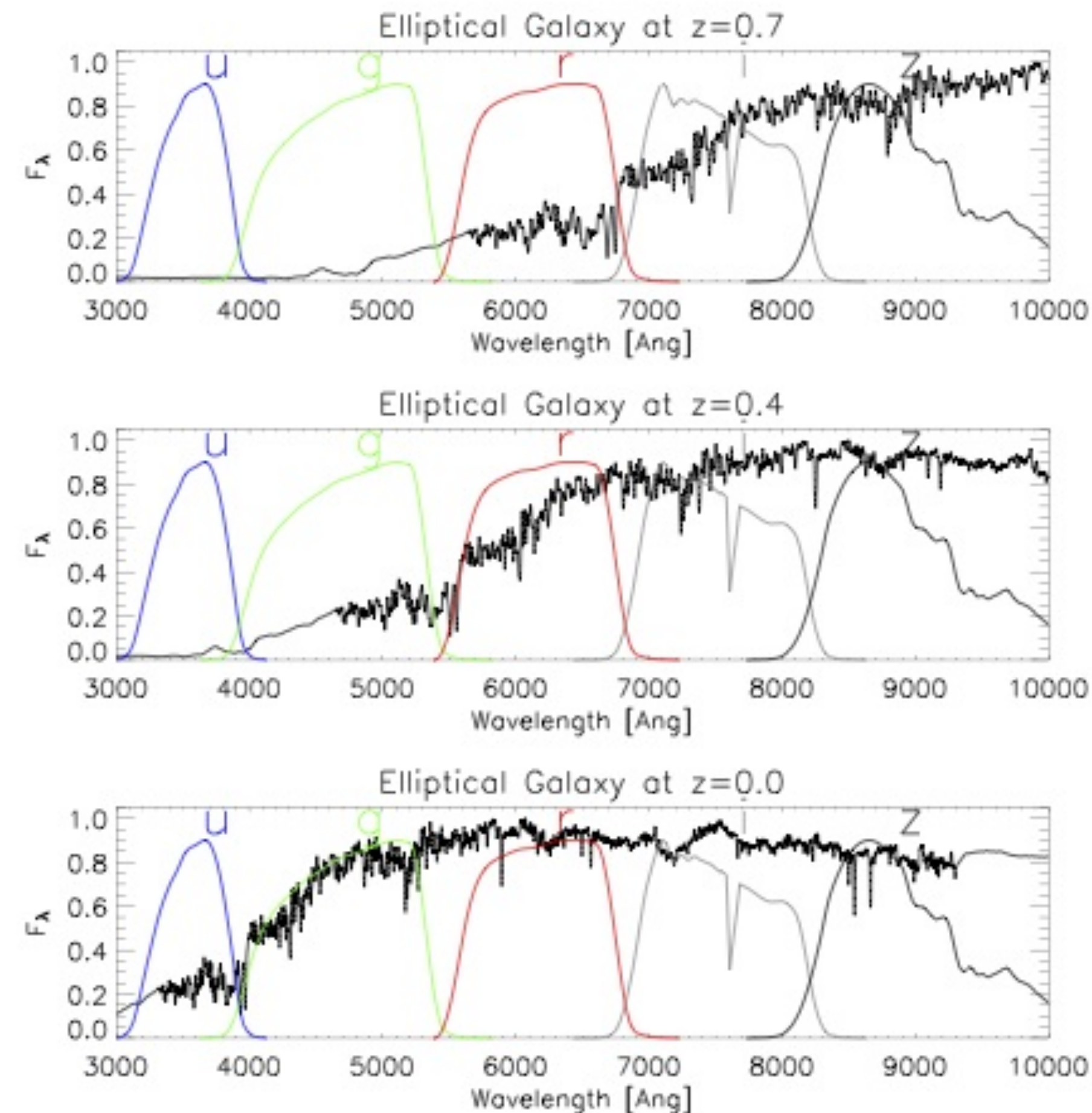
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e.g. photometric redshift distribution

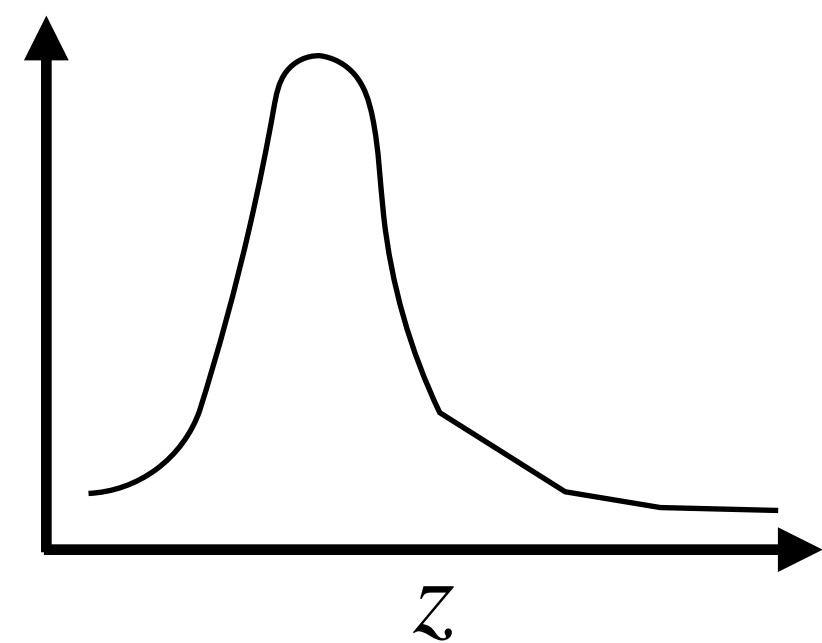
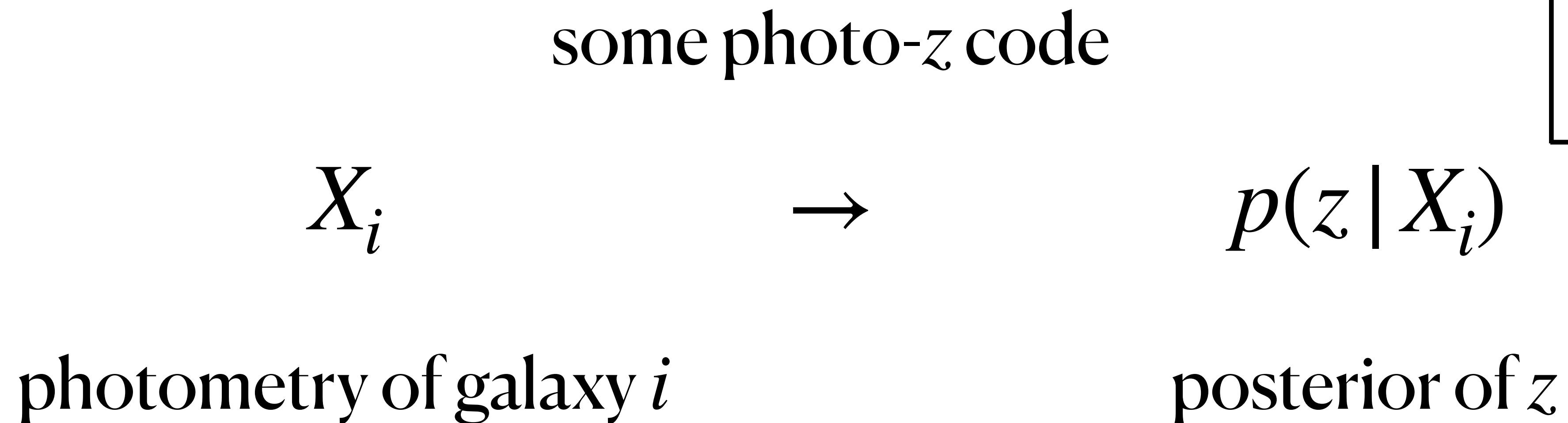


weak lensing cosmology: we want to measure how much matter
between us and source galaxies — we need to know the redshift
distribution, $n(z)$, of our sources based on photometric redshifts

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some photo- z code

X_i

$\rightarrow$

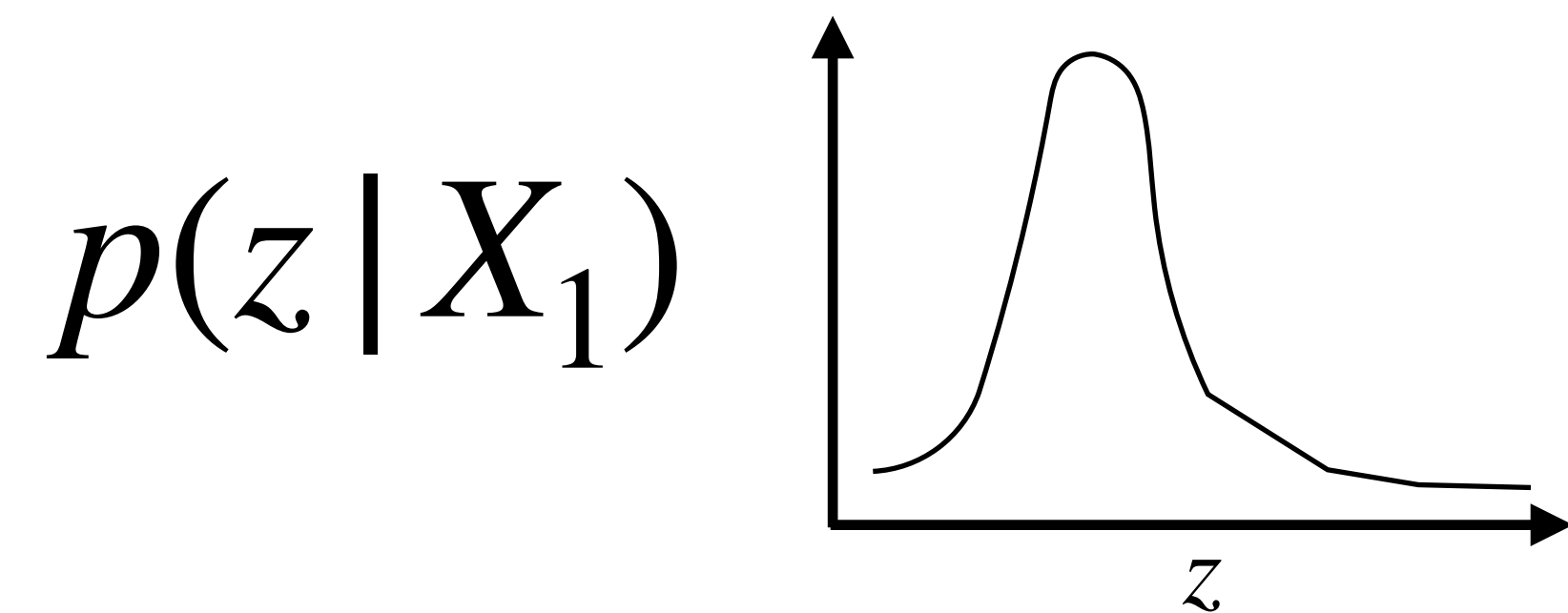
$p(z | X_i)$

photometry of galaxy i

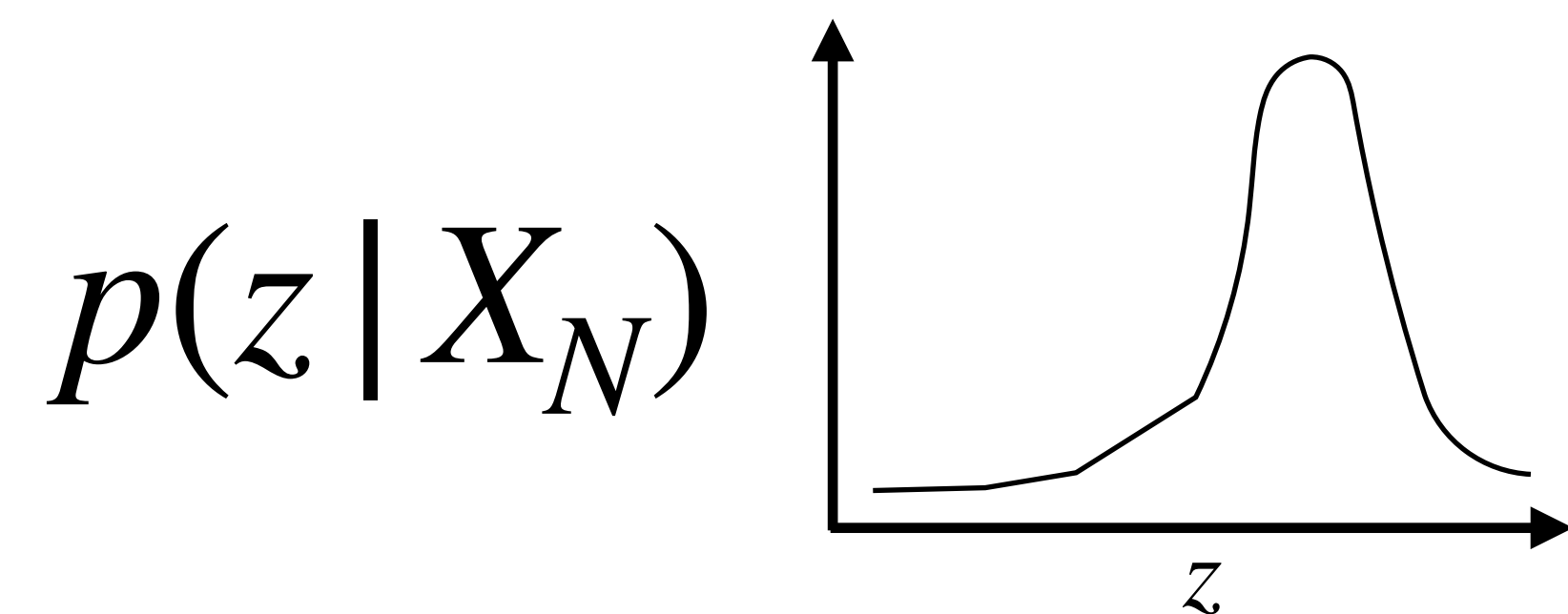
posterior of z

do this for N galaxies in your sample

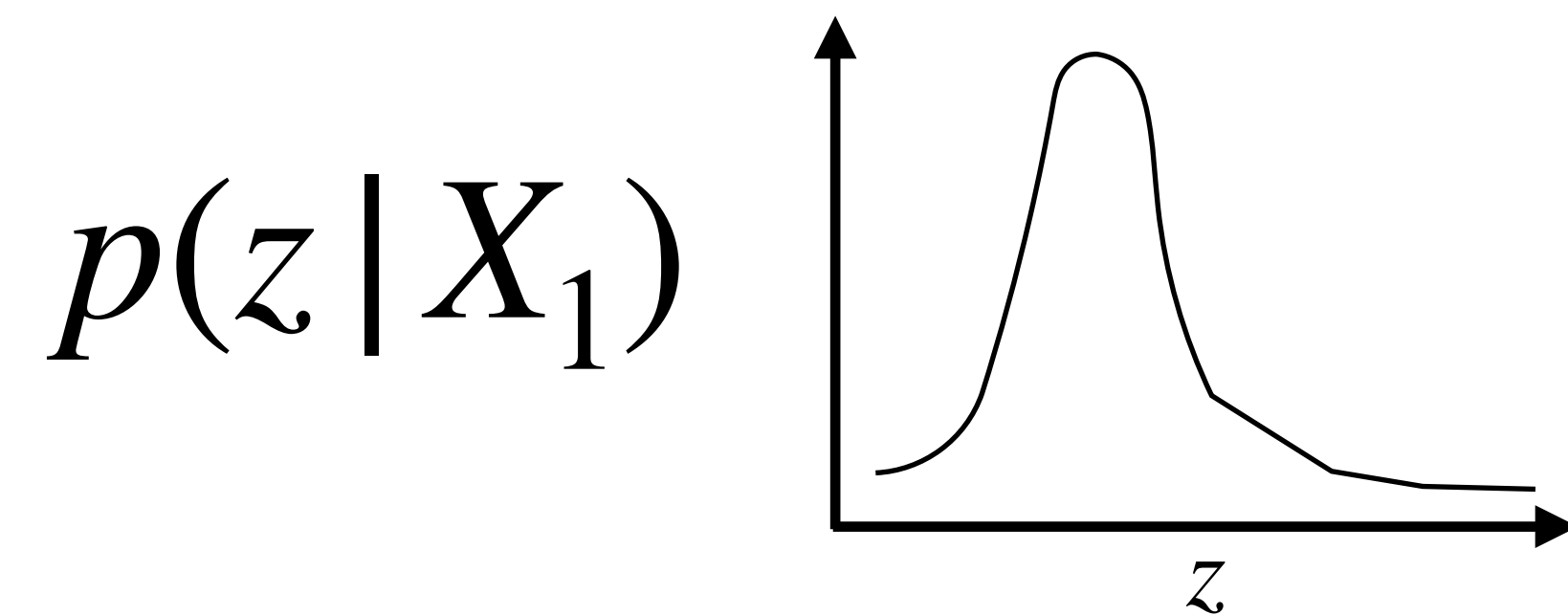
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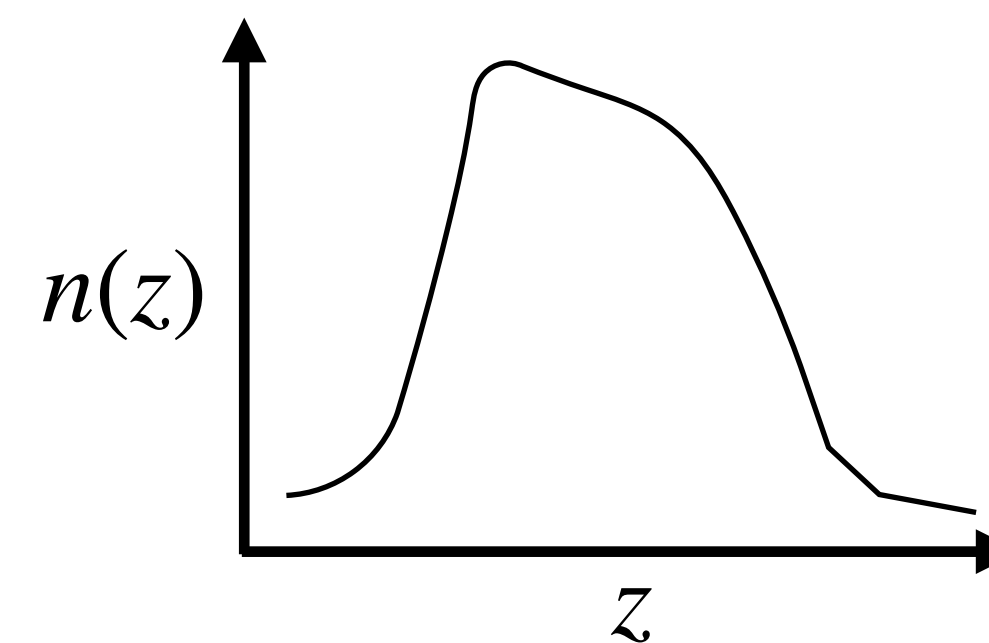
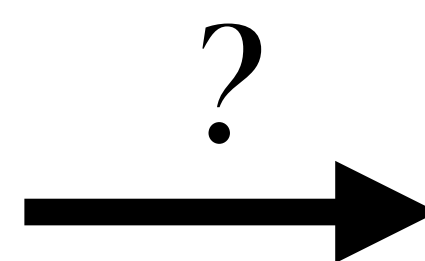
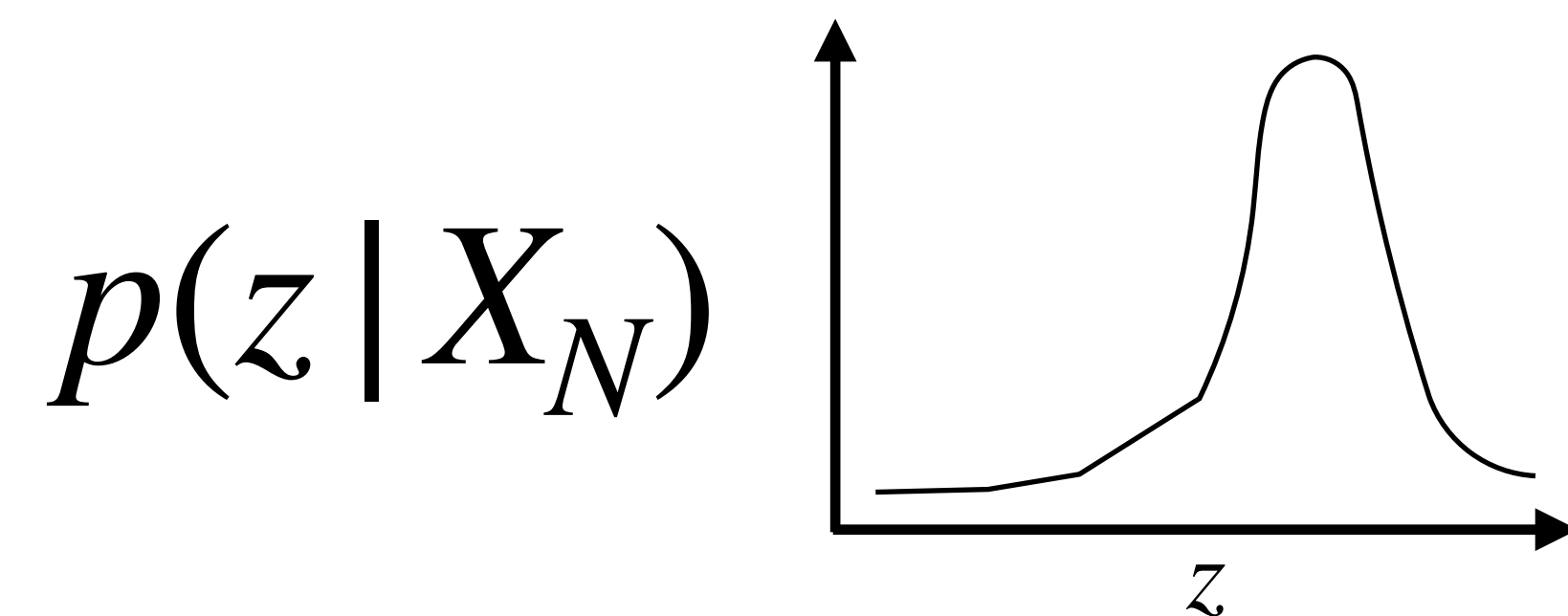
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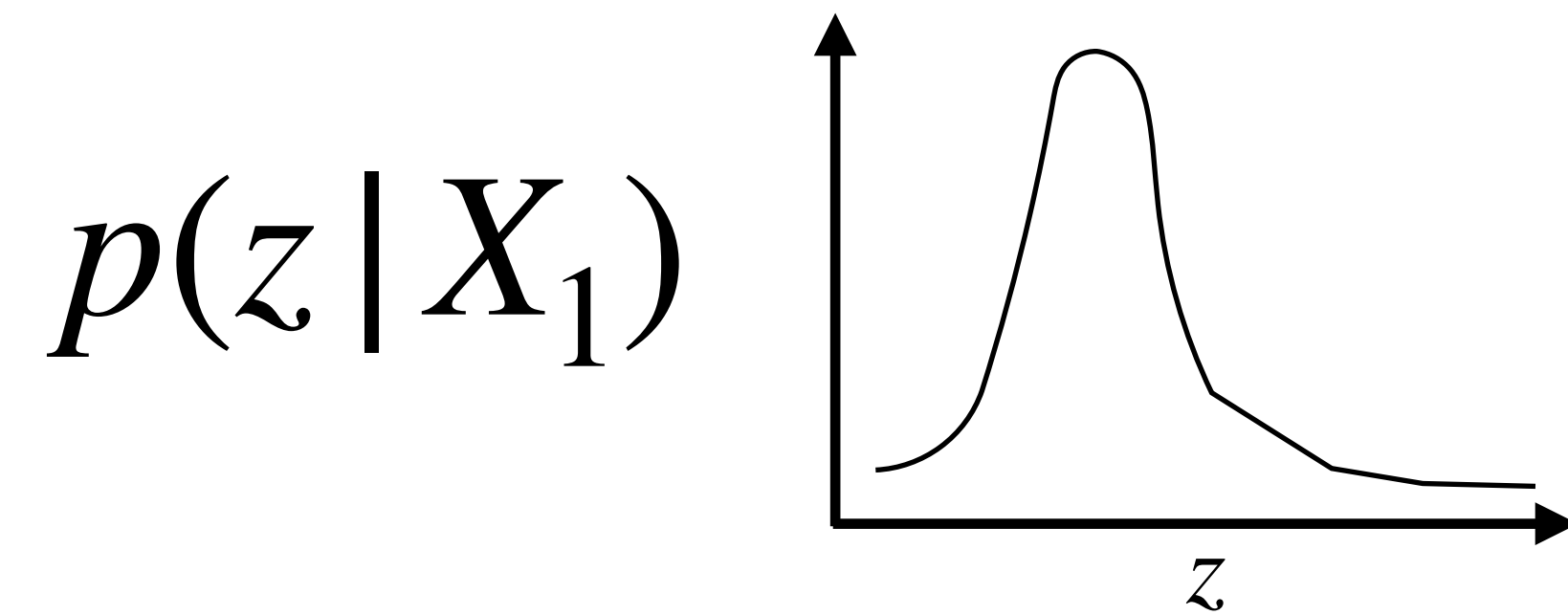
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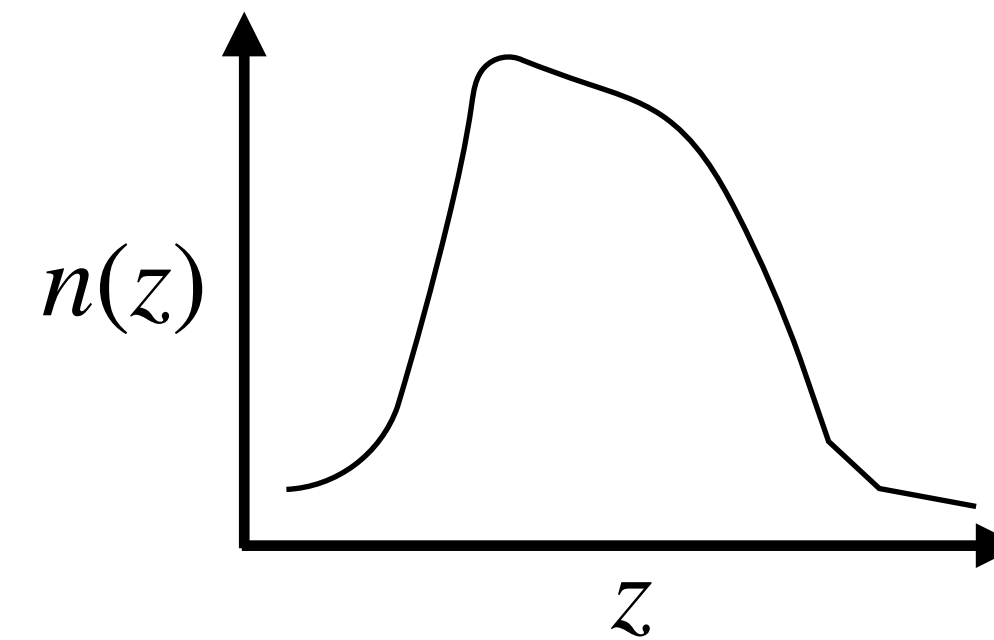
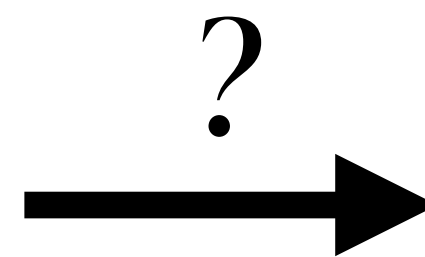
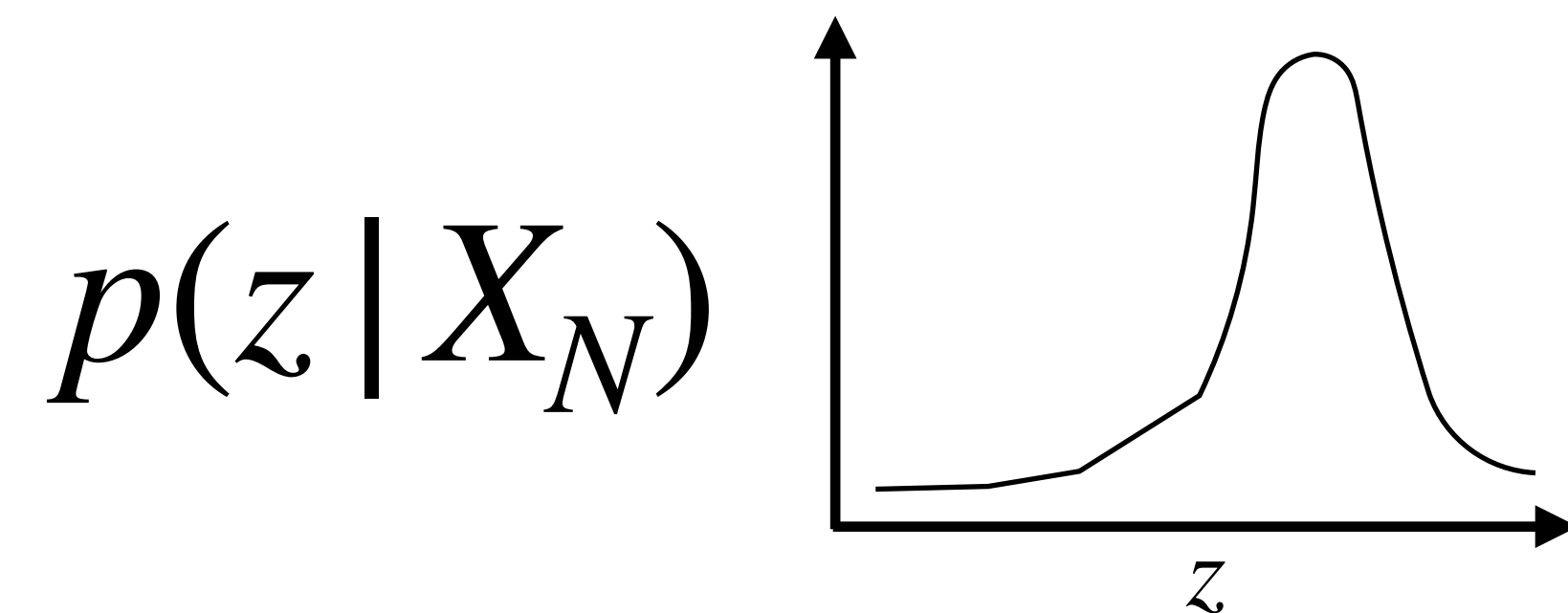
the redshift distribution $n(z)$

stacking? $n(z) = \frac{1}{N} \sum_i p(z | X_i)$

histogram the MAP? (*mode*)



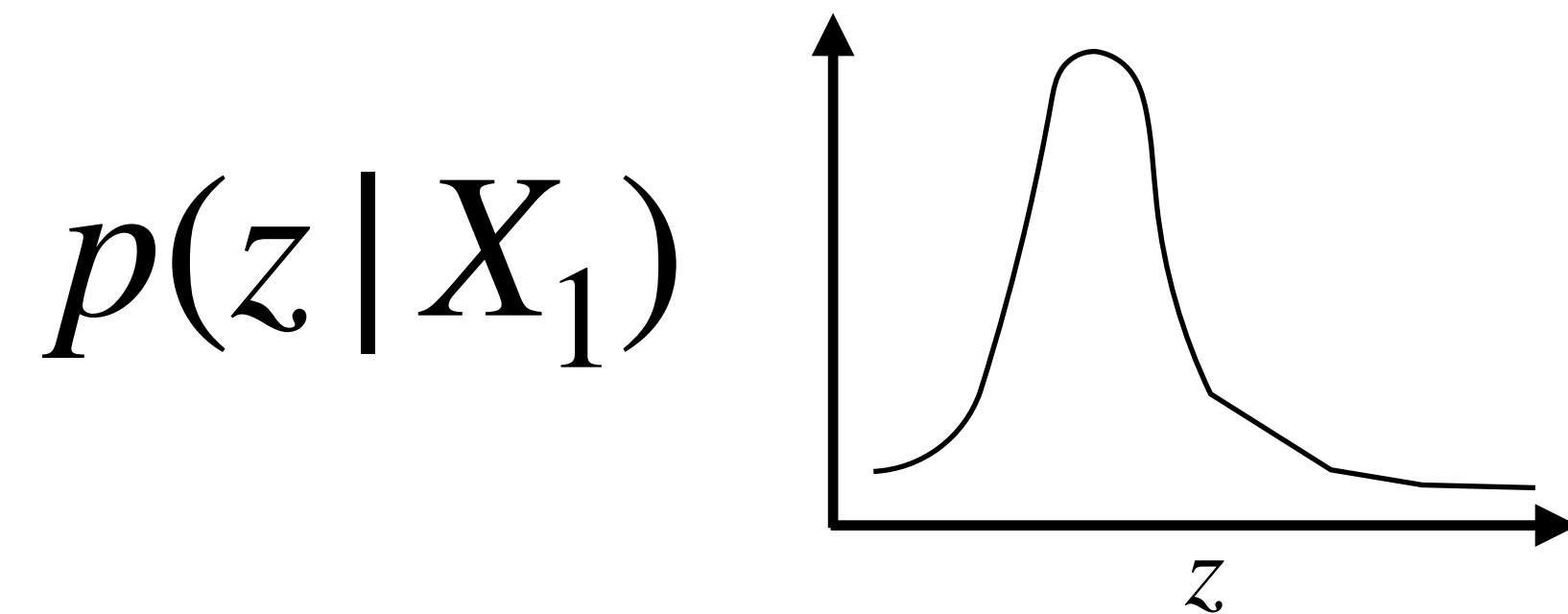
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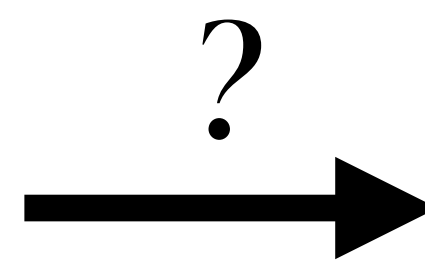
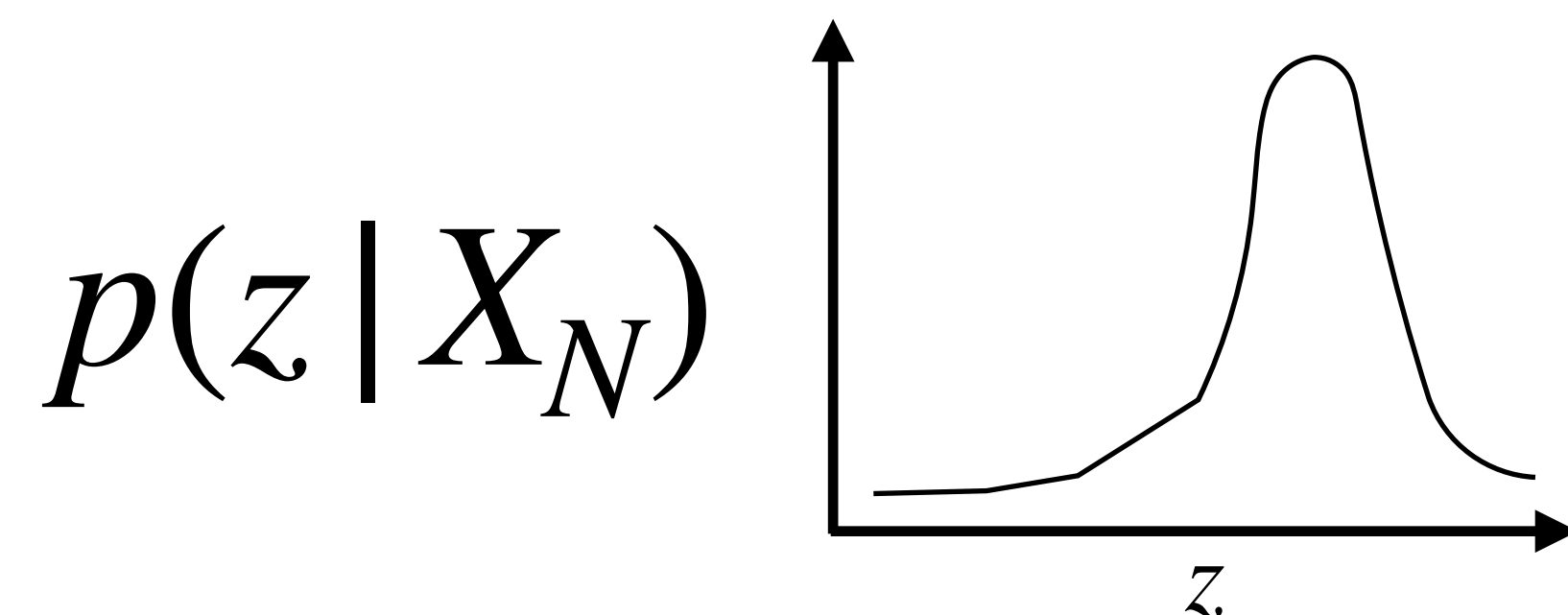
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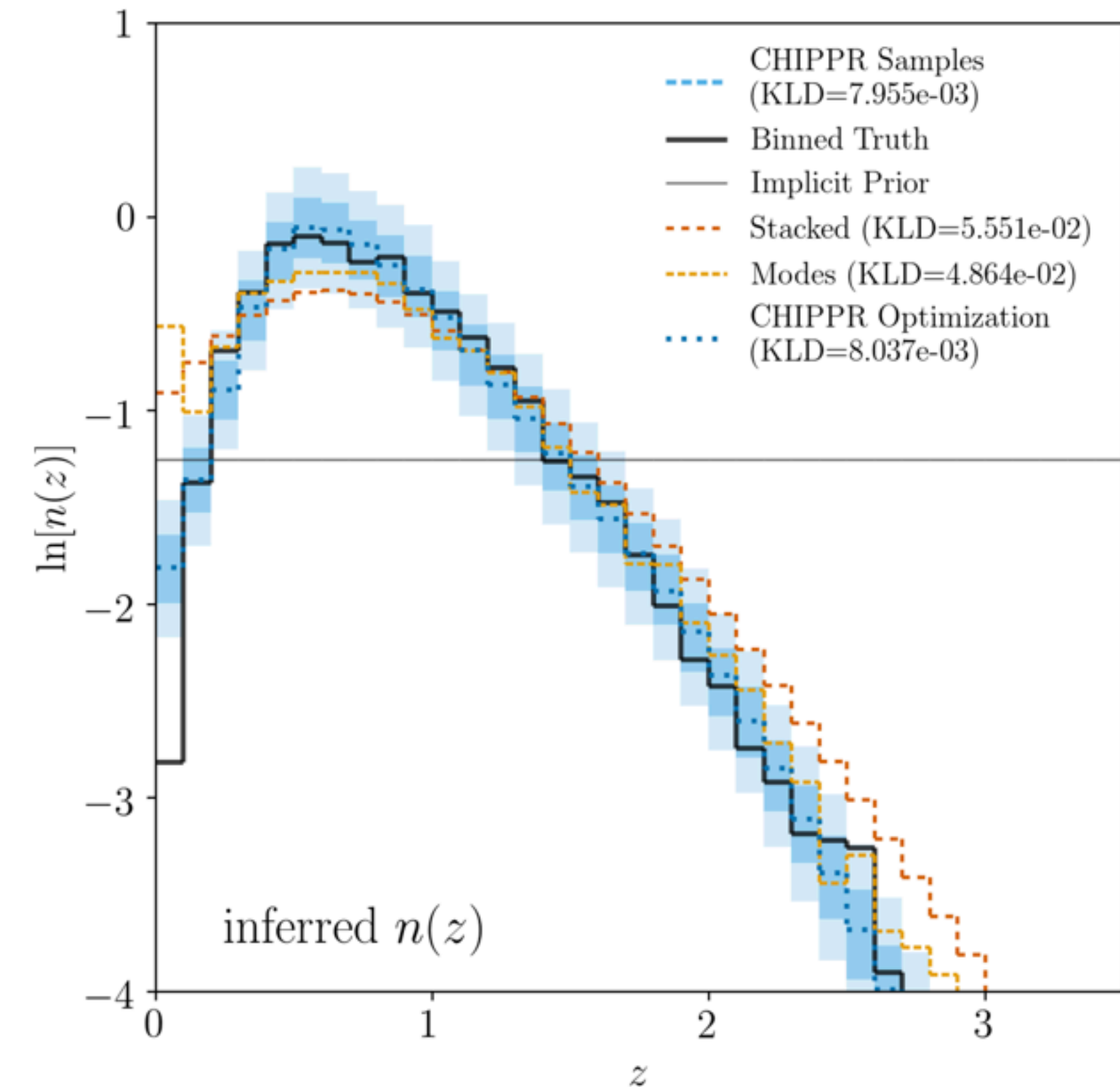
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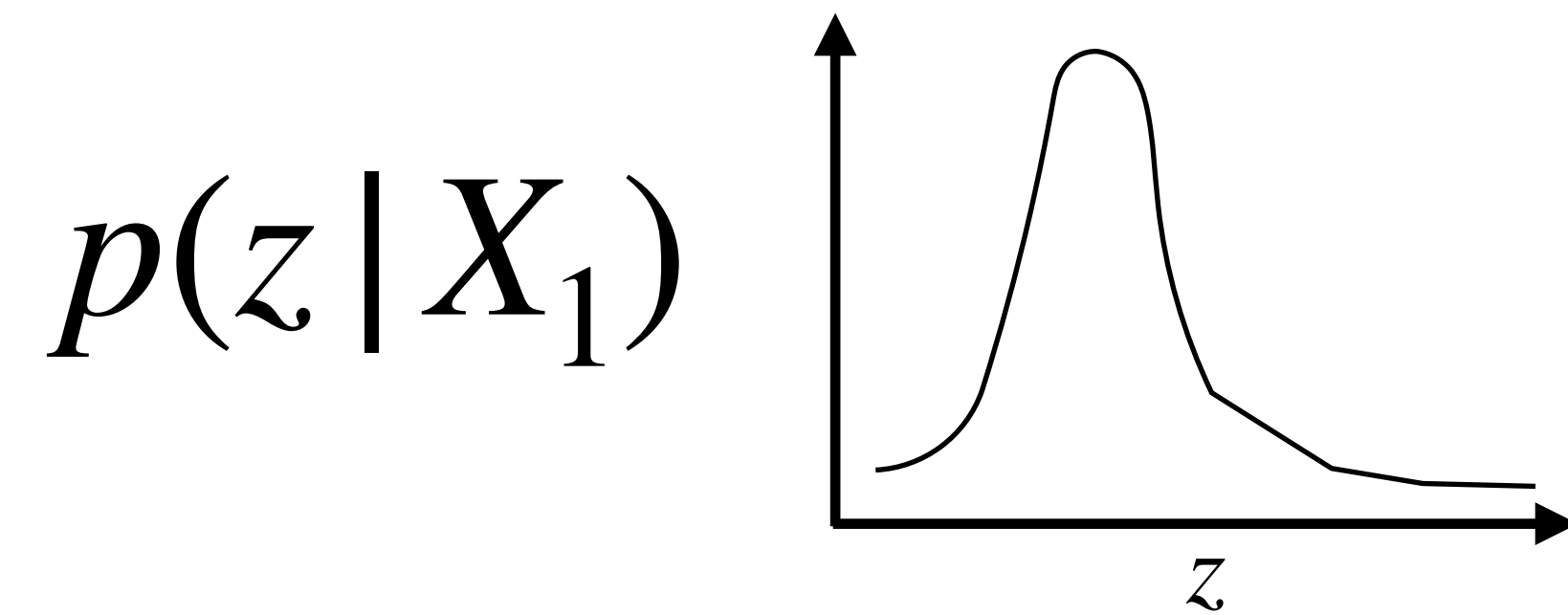


both are biased!

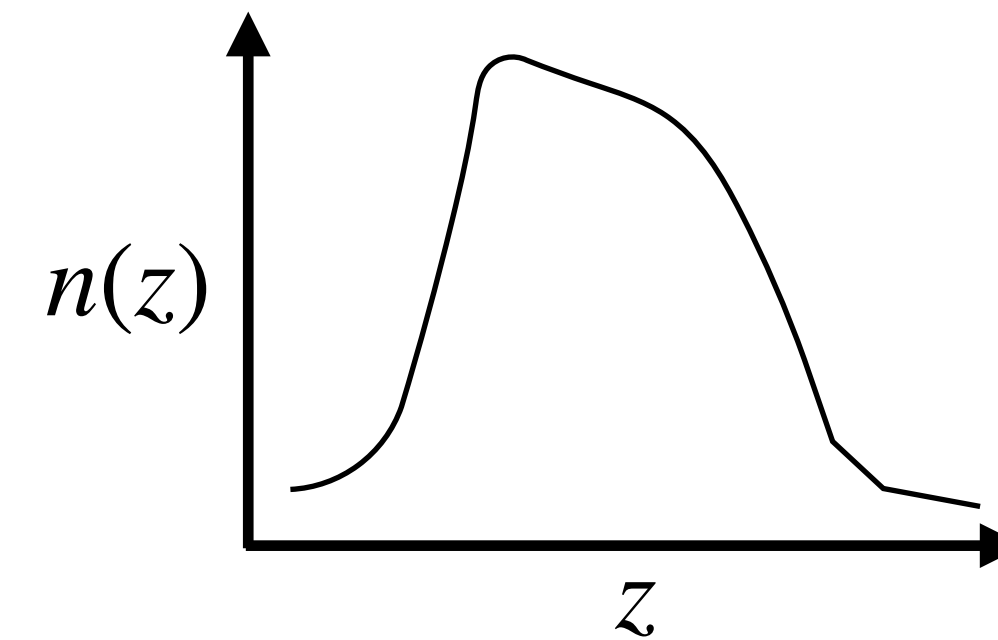
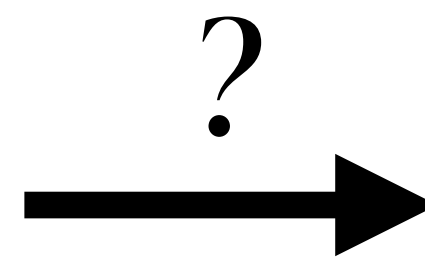
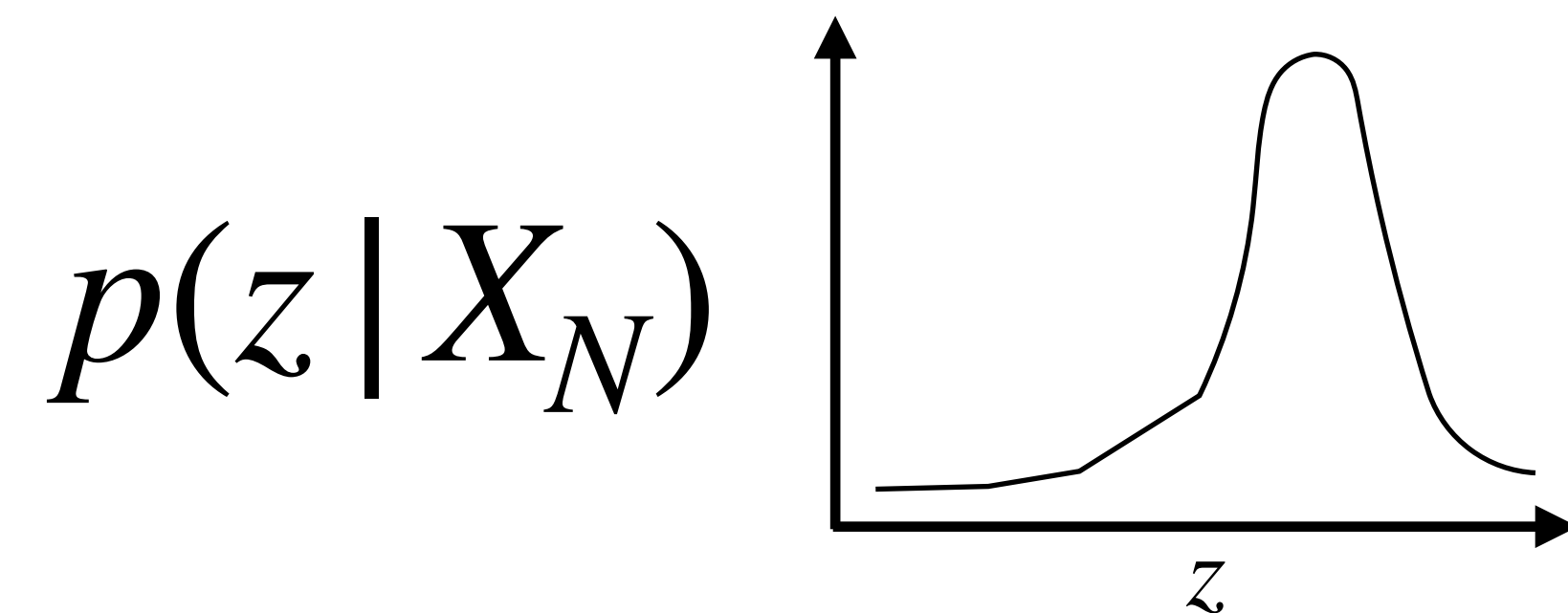


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that describe $n(z)$

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$$p(\phi \mid \{X_i\}_{i=1\dots N})$$

how can we infer $n(z)$ from the dataset $\{X_i\}_{i=1\dots N}$? —
 $p(\phi \mid \{X_i\}_{i=1\dots N})$

$$\propto p(\phi) p(\{X_i\} \mid \phi) \qquad \textit{Bayes' rule}$$

$$\propto p(\phi) \prod_i p(X_i \mid \phi) \qquad \textit{X_i's are independent}$$

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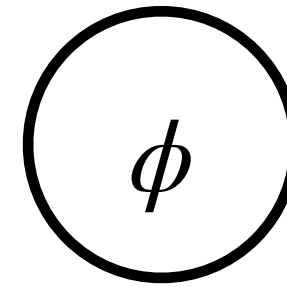
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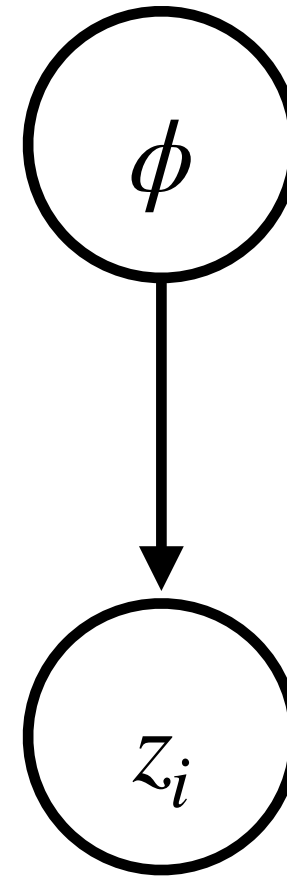
Hierarchical Bayes: forward model/data generation process



*hyperparameter
describing $n(z)$*

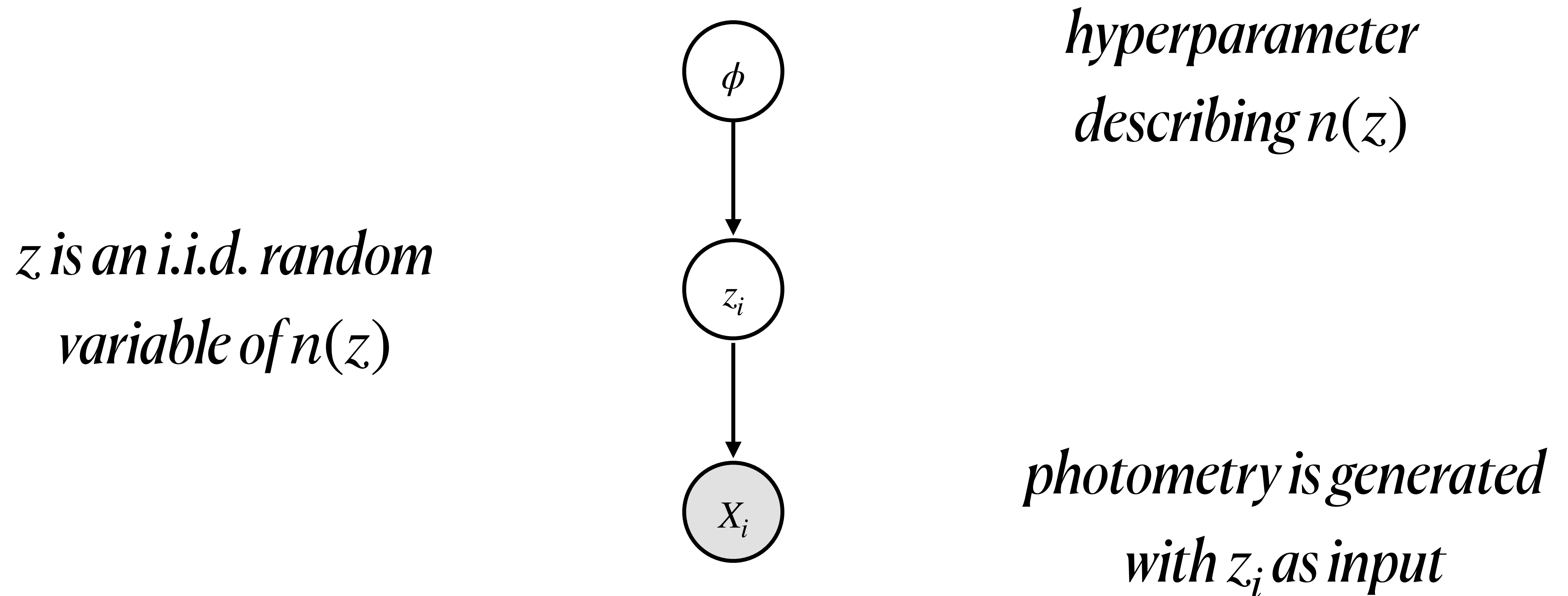
Hierarchical Bayes: forward model/data generation process

*z is an i.i.d. random
variable of $n(z)$*

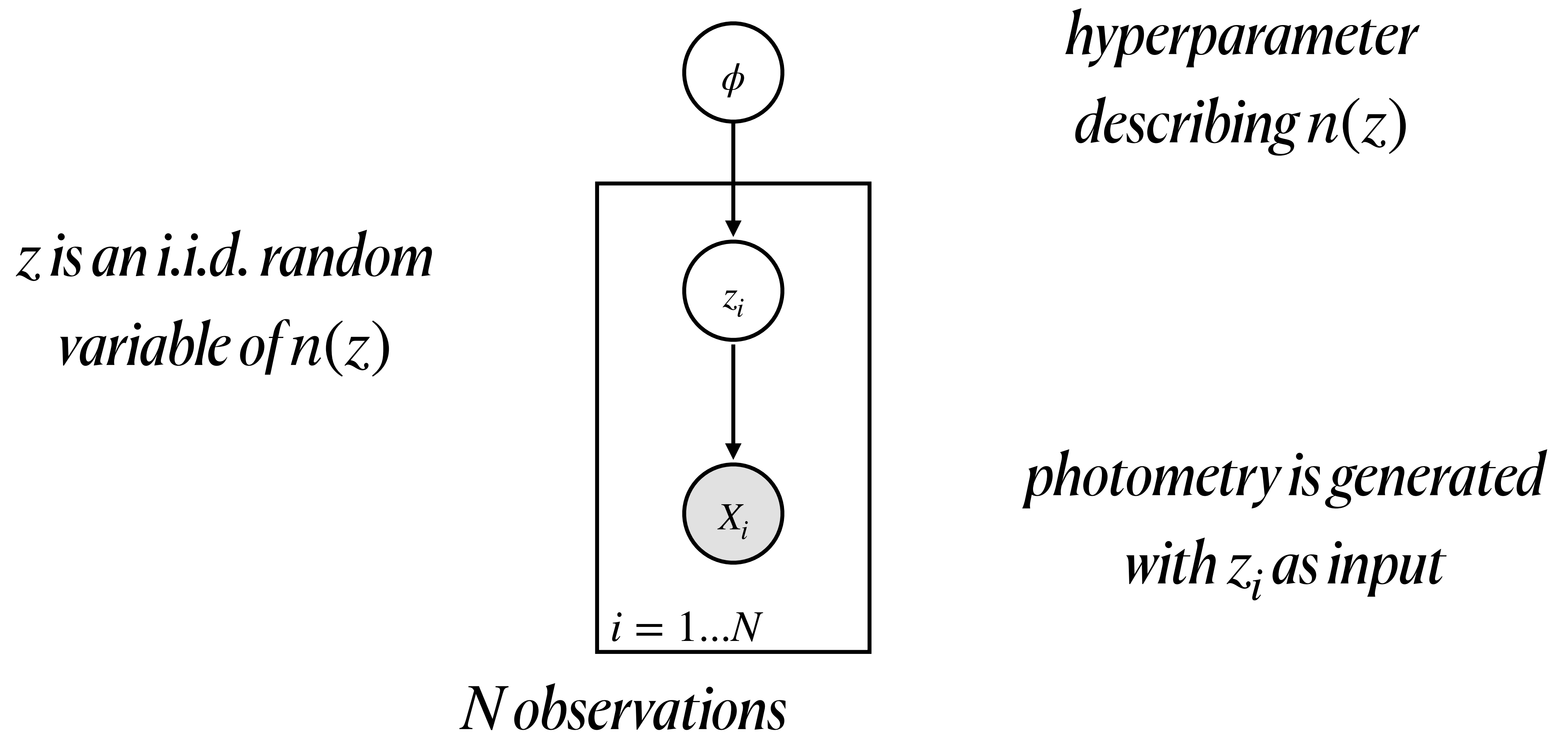


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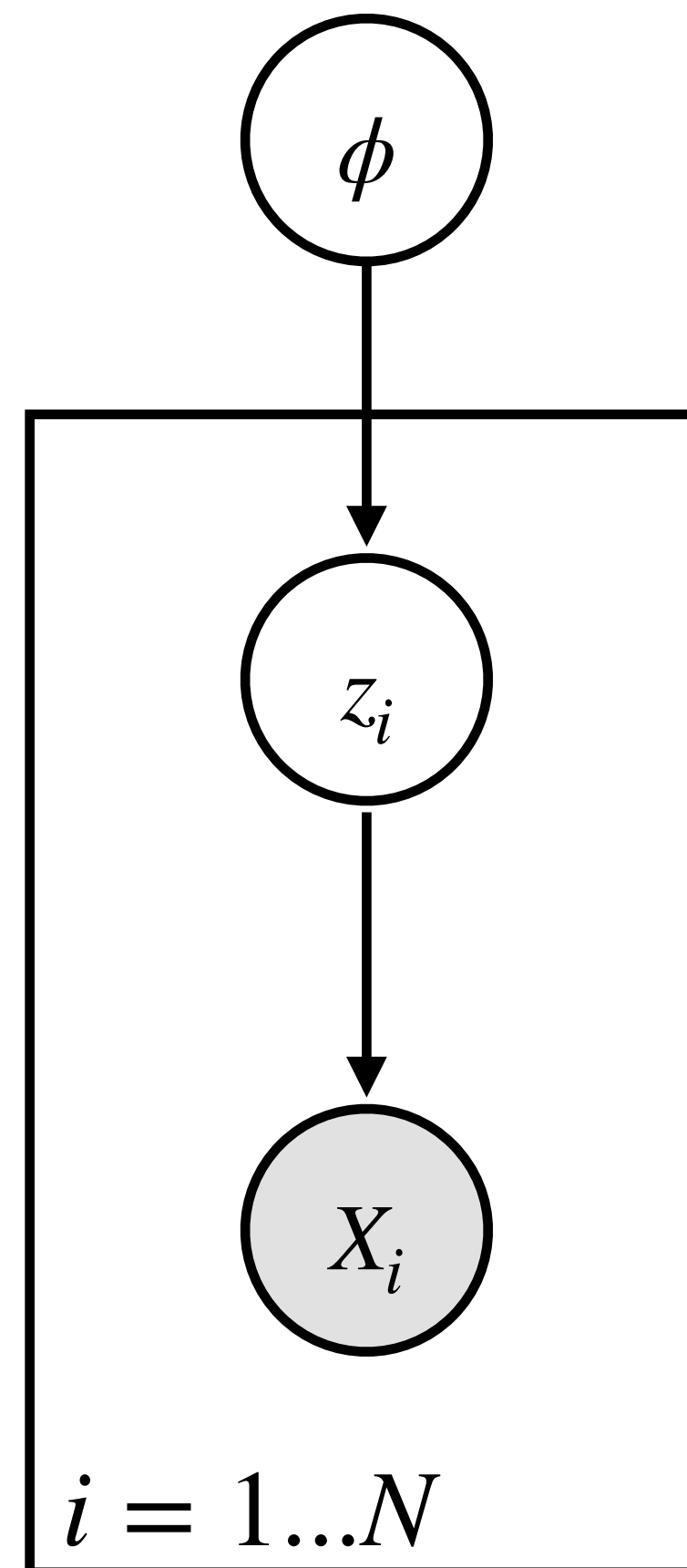
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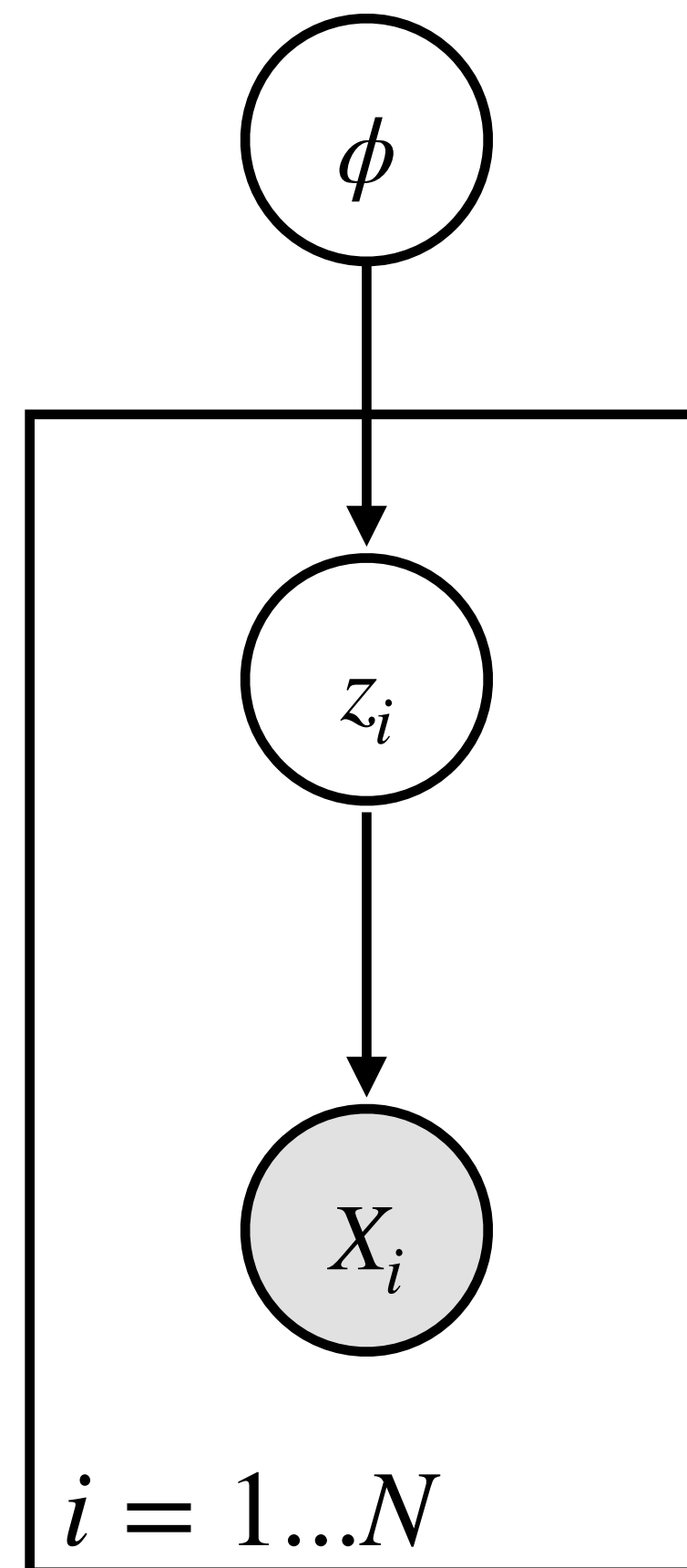
Hierarchical Bayes: forward model/data generation process



probabilistic graphical models (PGM) to represent hierarchical problems — *Directed Acyclic Graph (DAG)*

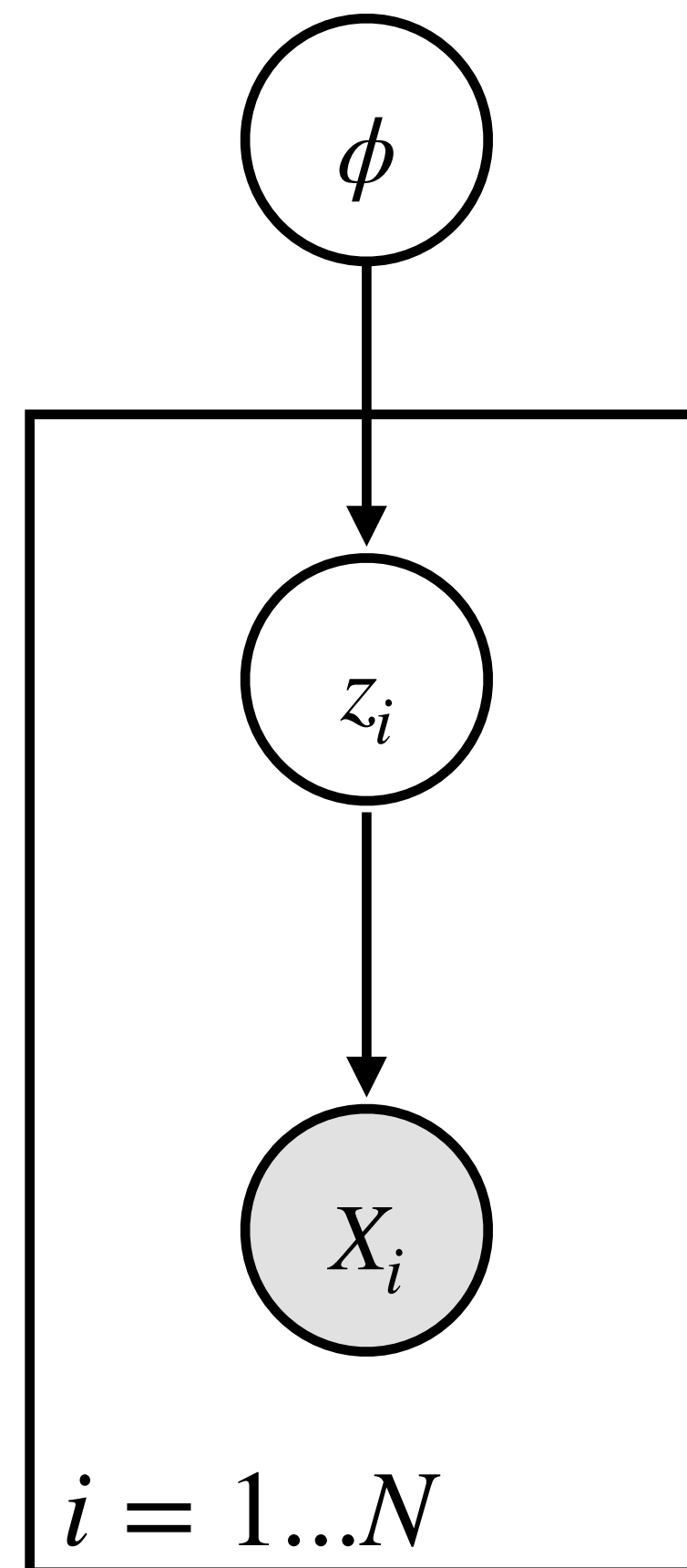


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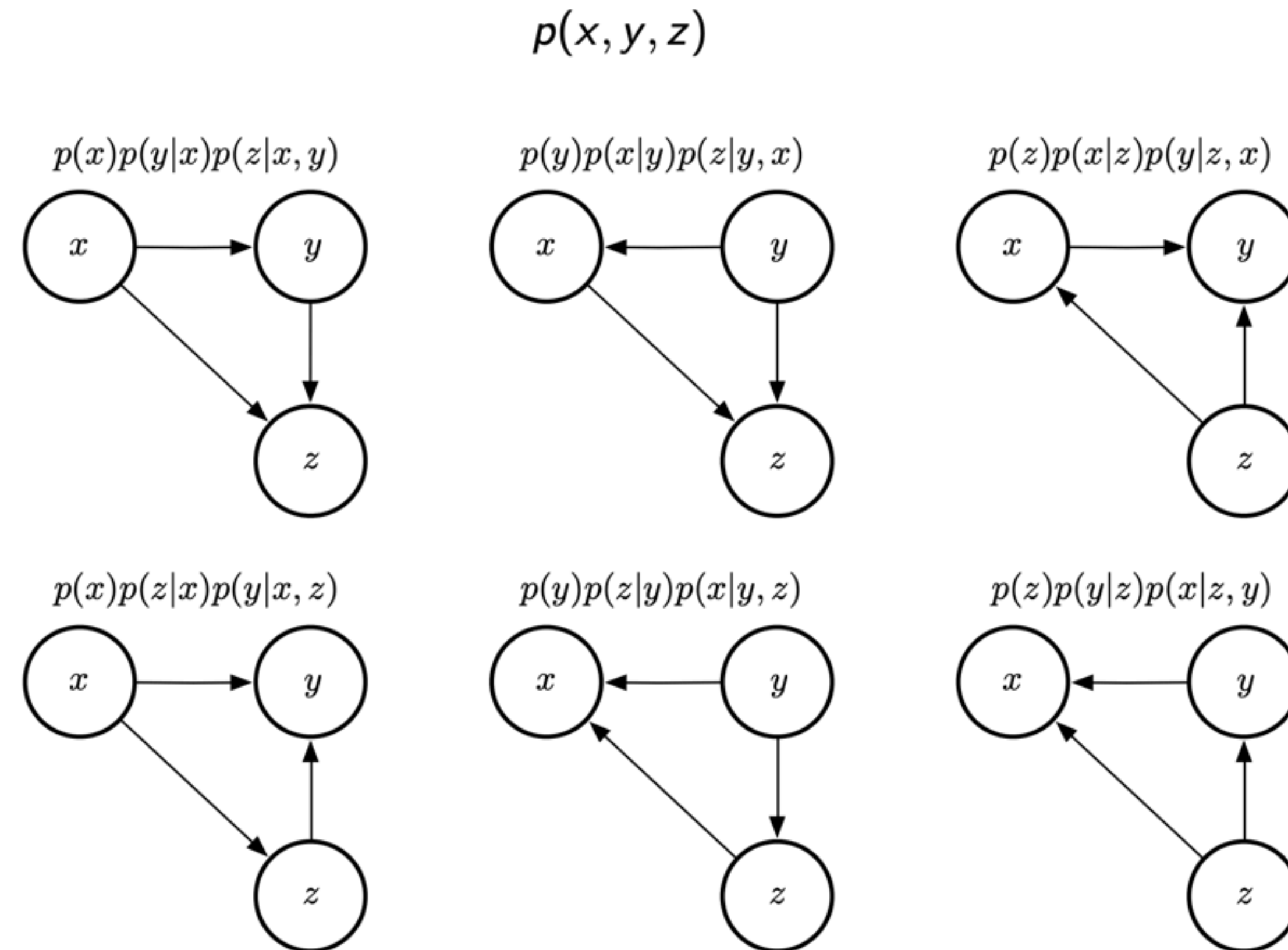
circles (nodes) represent random variables; shaded is observed
rectangles (plates) represent multiple draws

probabilistic graphical models (PGM) to represent hierarchical problems — *Directed Acyclic Graph (DAG)*



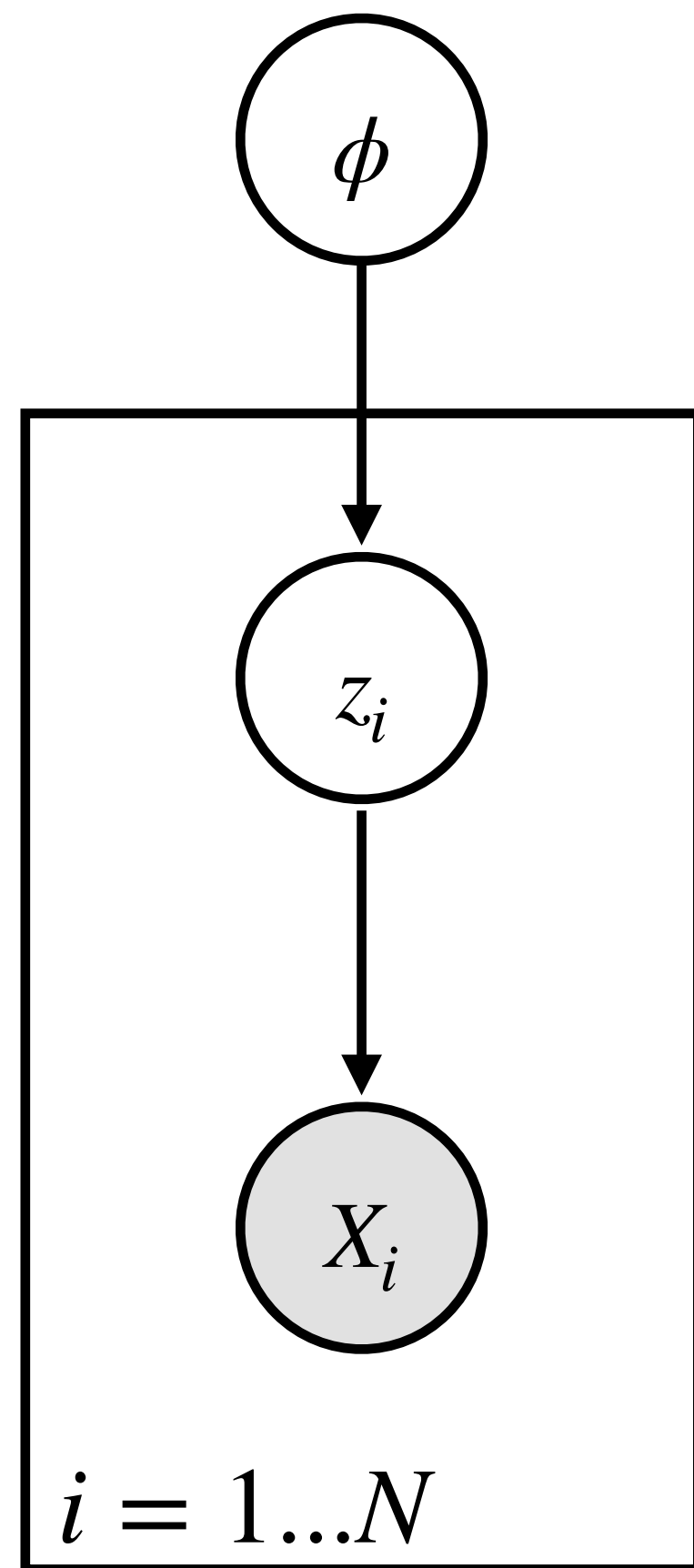
edges or their absence represent conditional dependence or independence

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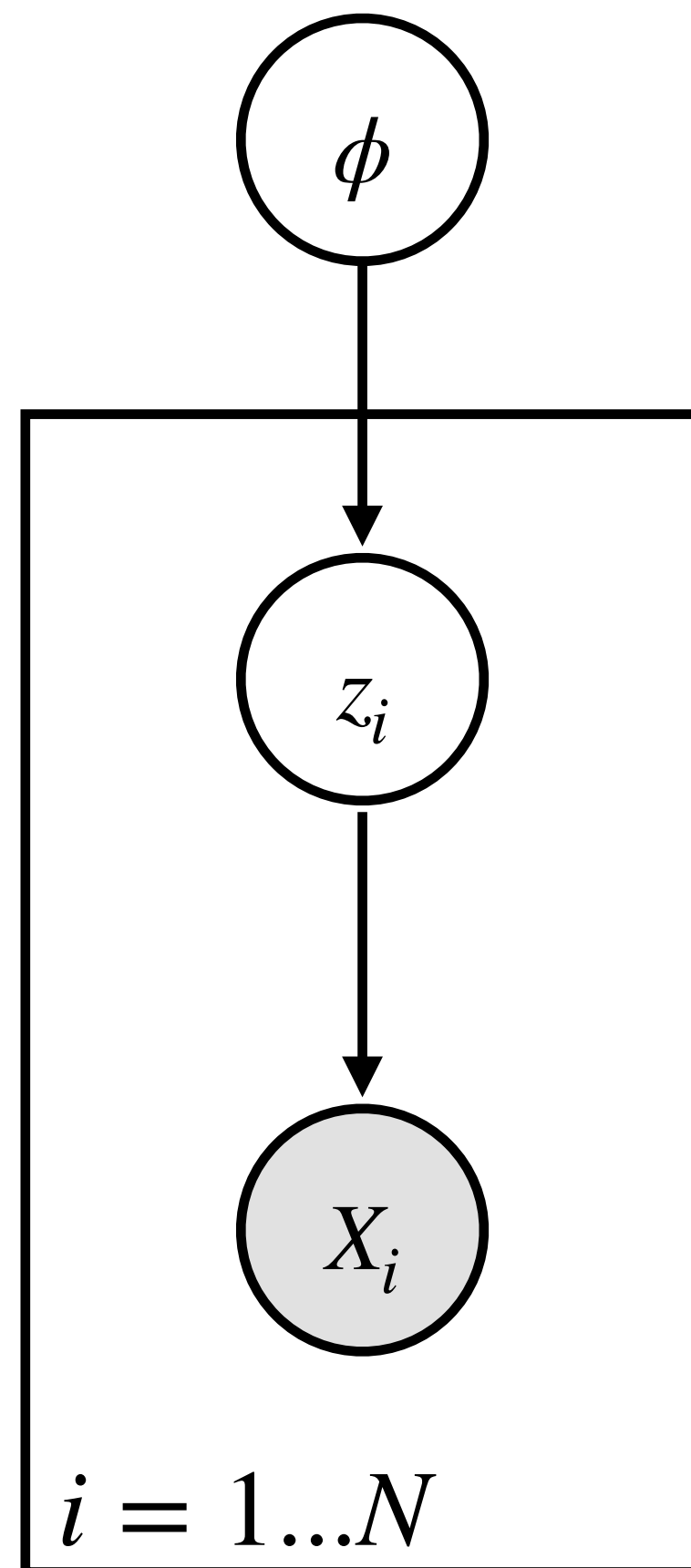


graph maps to a joint density!

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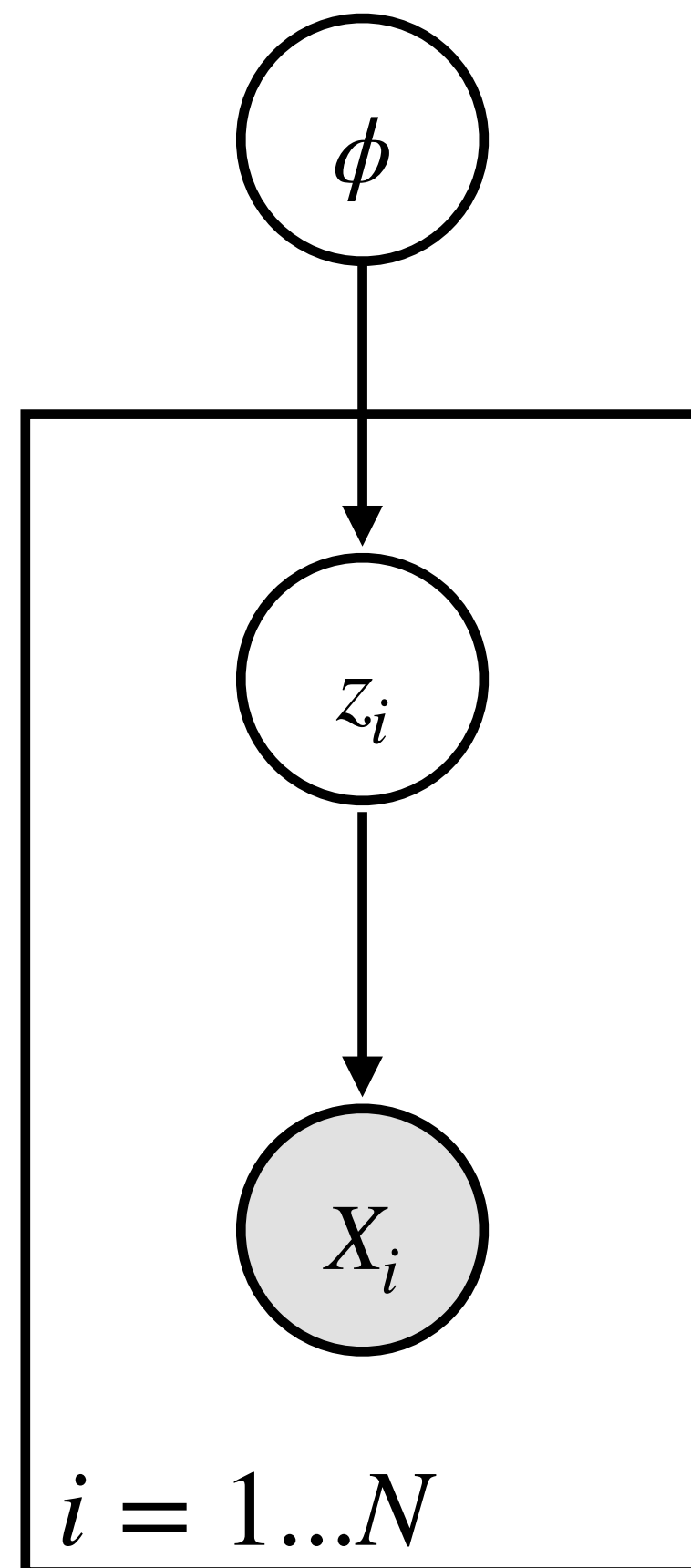
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but we want $p(\phi | \{X_i\})!$



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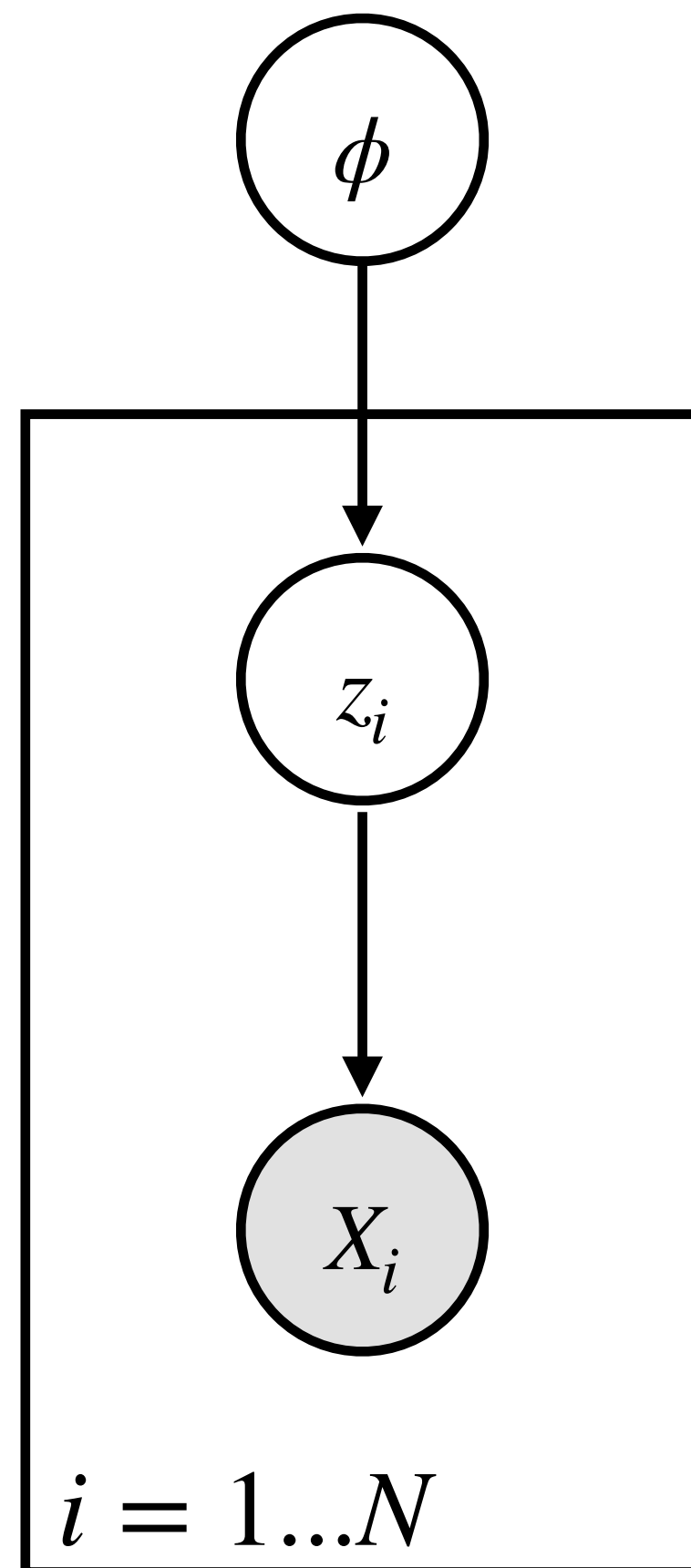


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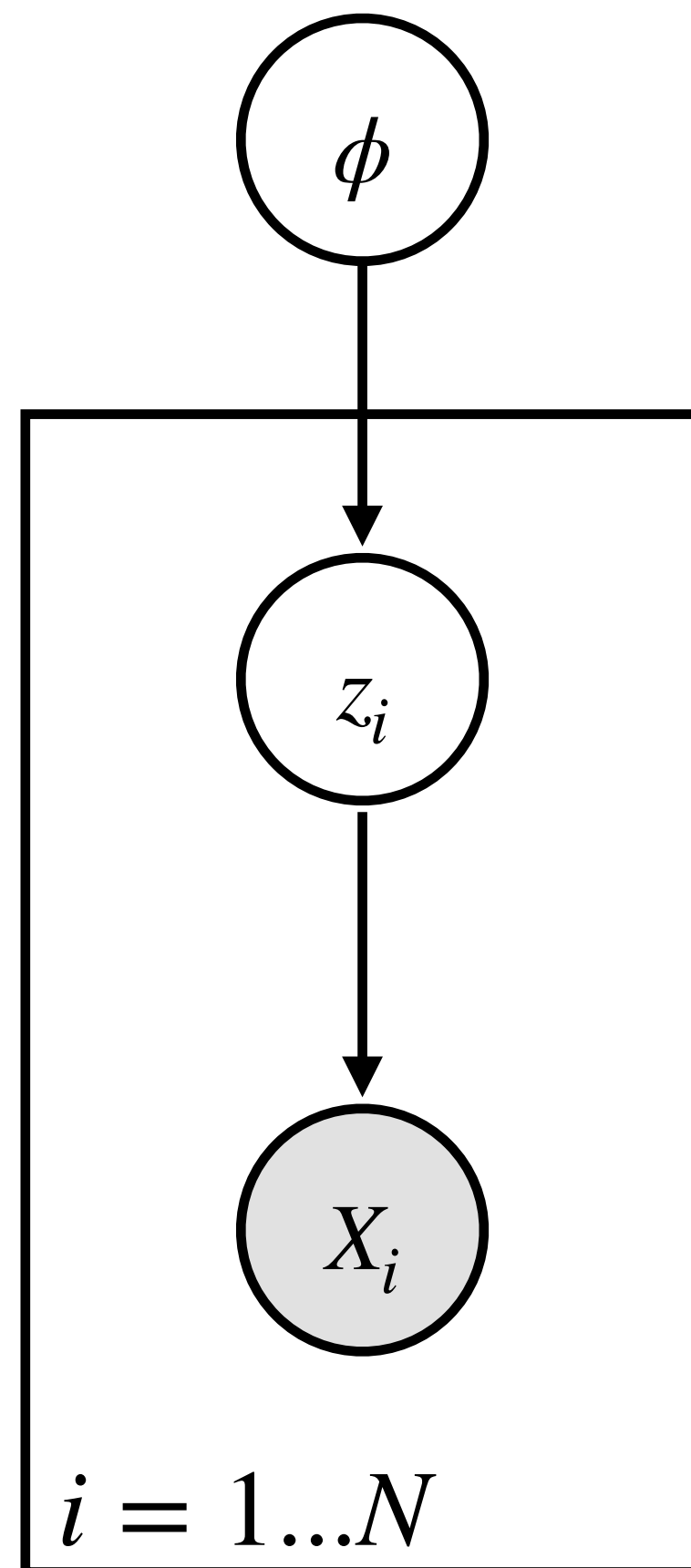
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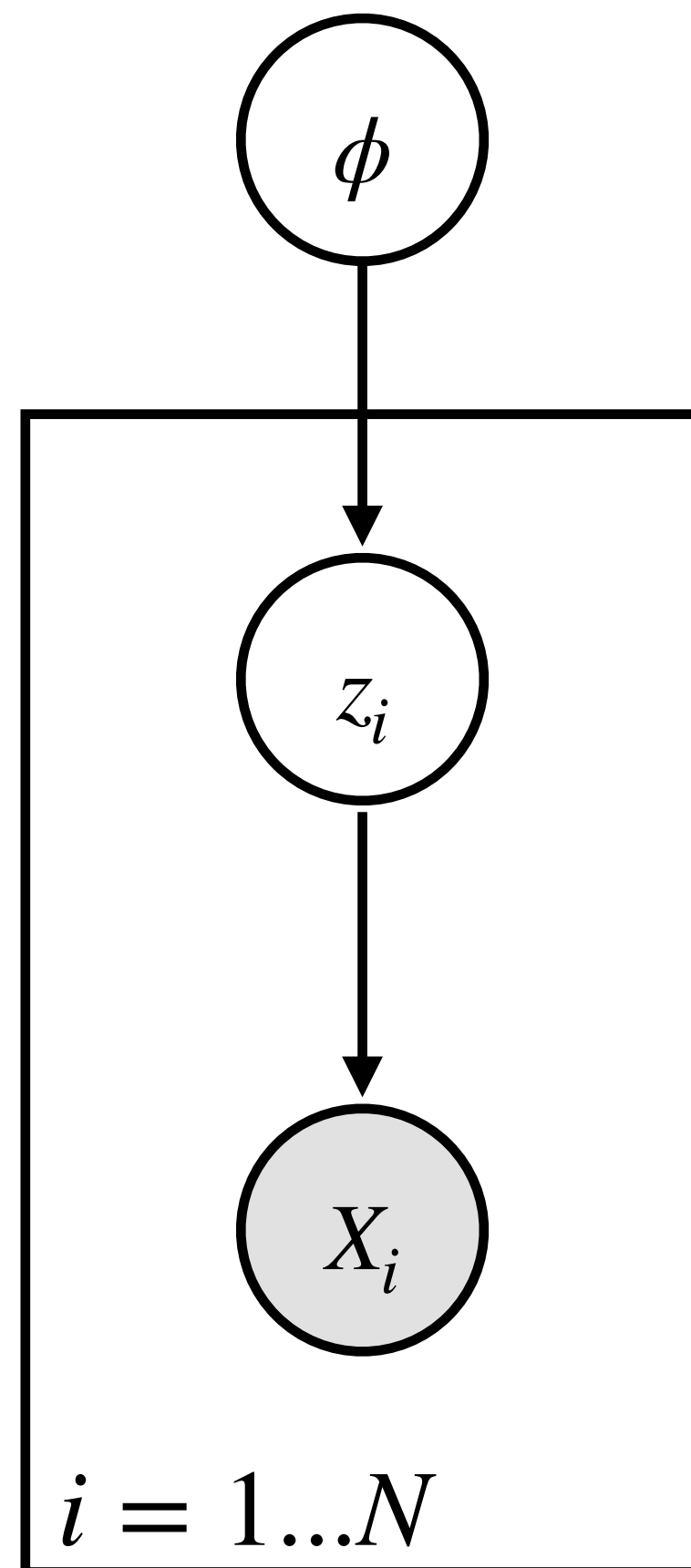
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$z_0 \dots z_N$ are **conditionally independent**

$$p(\phi, \{z_i\}, \{X_i\}) = p(\phi) \prod_i p(z_i | \phi) p(X_i | z_i)$$

but we want $p(\phi | \{X_i\})$!



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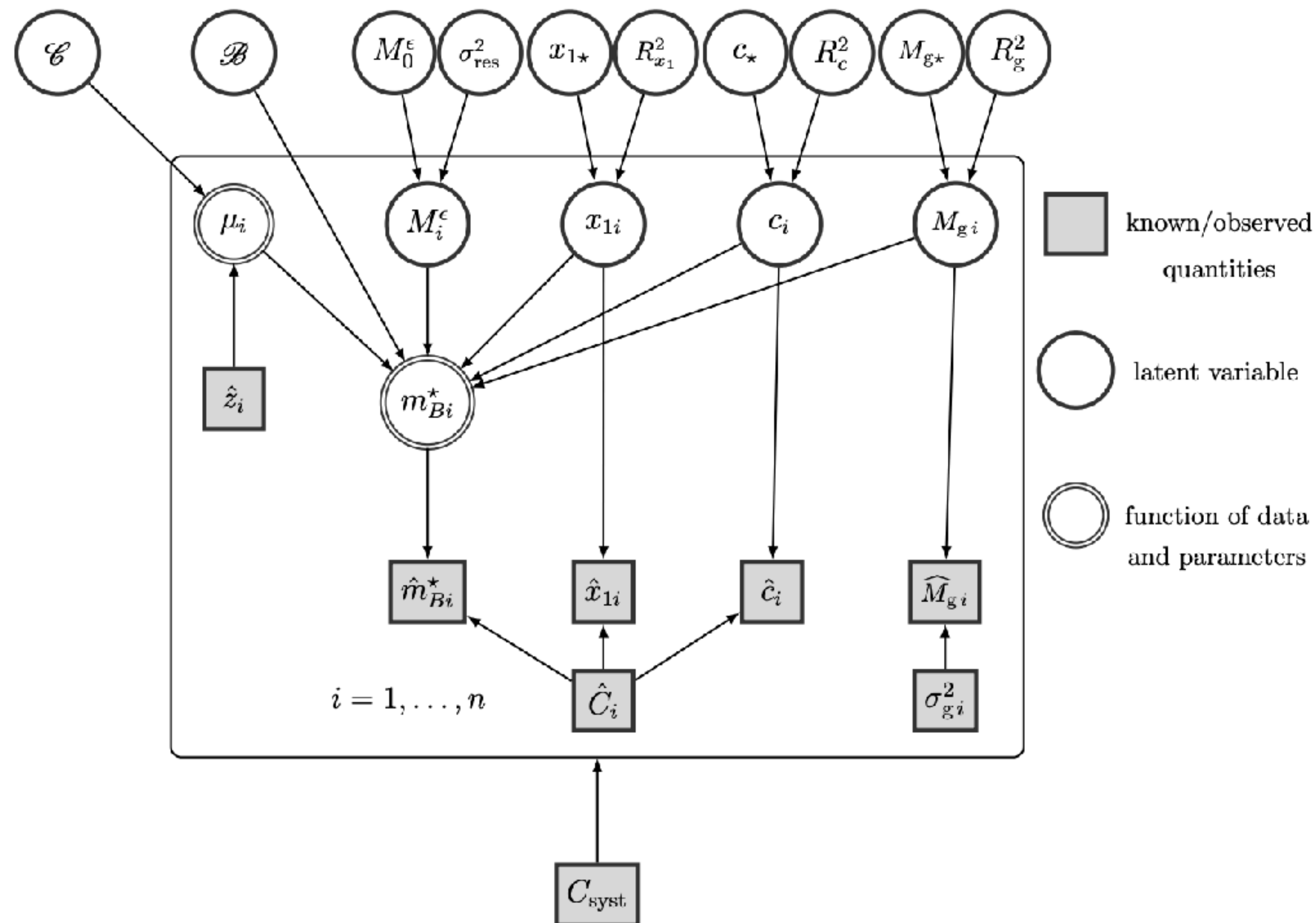
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same expression as earlier

hierarchical Bayesian modeling can get complicated



when should you use hierarchical Bayesian modeling?

“hierarchical bayesian modeling is a great way to spend three times longer for the same results”

when your data generation process is hierarchical:
hyperparameters $\rightarrow$ latent parameters $\rightarrow$ observables

when individual posteriors are broad and have large uncertainties
(low signal-to-noise regimes)

in-class: population inference using hierarchical Bayes

ast384c/inclass/lec9_hierarchical/hi_bayes.ipynb